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Assignment

Bachelor of Computer Applications

Semester III

Database Management System

Topic: ER Diagram and Design and Relational Schema

BY

Name: M Blessy

Registration Number: 2511021240067

DEPARTMENT OF COMPUTER APPLICATIONS

Alliance university

Chikkahagade cross, Anekal, Bangaluru 562106

OBJECTIVE

To design a complete Entity Relationship (ER) Diagram for the Hospital Management System and convert it into a Relational Schema by identifying entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

ENTITIES AND ATTRIBUTES

Entity 1: DEPARTMENT

Column	Key
department_id	Primary Key
department_name	-
floor_no	-
phone_no	-
head_doctor_id	Foreign Key

Entity 2: DOCTOR

Column	Key
doctor_id	Primary Key
name	-
specialization	-
department_id	Foreign Key
phone_no	-
email	-
experience_years	-
consultation_fee	-

Entity 3: PATIENT

Column	Key
patient_id	Primary Key
name	-
dob	-
gender	-
phone_no	-
address	-
blood_group	-
emergency_contact	-

Entity 4: APPOINTMENT

Column	Key
appointment_id	Primary Key
patient_id	Foreign Key
doctor_id	Foreign Key
appointment_date	-
appointment_time	-
status	-
reason	-

Entity 5: ROOM

Column	Key
room_id	Primary Key
room_number	-
room_type	-
floor_no	-
daily_charge	-
status	-

Entity 6: ADMISSION

Column	Key
admission_id	Primary Key
patient_id	Foreign Key
room_id	Foreign Key
doctor_id	Foreign Key
admission_date	-
discharge_date	-
admission_reason	-
status	-

Entity 7: PRESCRIPTION

Column	Key
prescription_id	Primary Key
appointment_id	Foreign Key
patient_id	Foreign Key
doctor_id	Foreign Key
date_issued	-
notes	-

Entity 8: MEDICINE

Column	Key
medicine_id	Primary Key
medicine_name	-
manufacturer	-
unit_price	-
stock_quantity	-

Entity 9: PRESCRIPTION_DETAIL

Column	Key
prescription_detail_id	Primary Key
prescription_id	Foreign Key
medicine_id	Foreign Key
dosage	-
duration_days	-
quantity	-

Entity 10: LAB_TEST

Column	Key
test_id	Primary Key
patient_id	Foreign Key
doctor_id	Foreign Key
test_name	-
test_date	-
result	-
status	-

Entity 11: BILL

Column	Key
bill_id	Primary Key
patient_id	Foreign Key
appointment_id	Foreign Key (nullable)
admission_id	Foreign Key (nullable)
bill_date	-
total_amount	-
payment_status	-
payment_mode	-

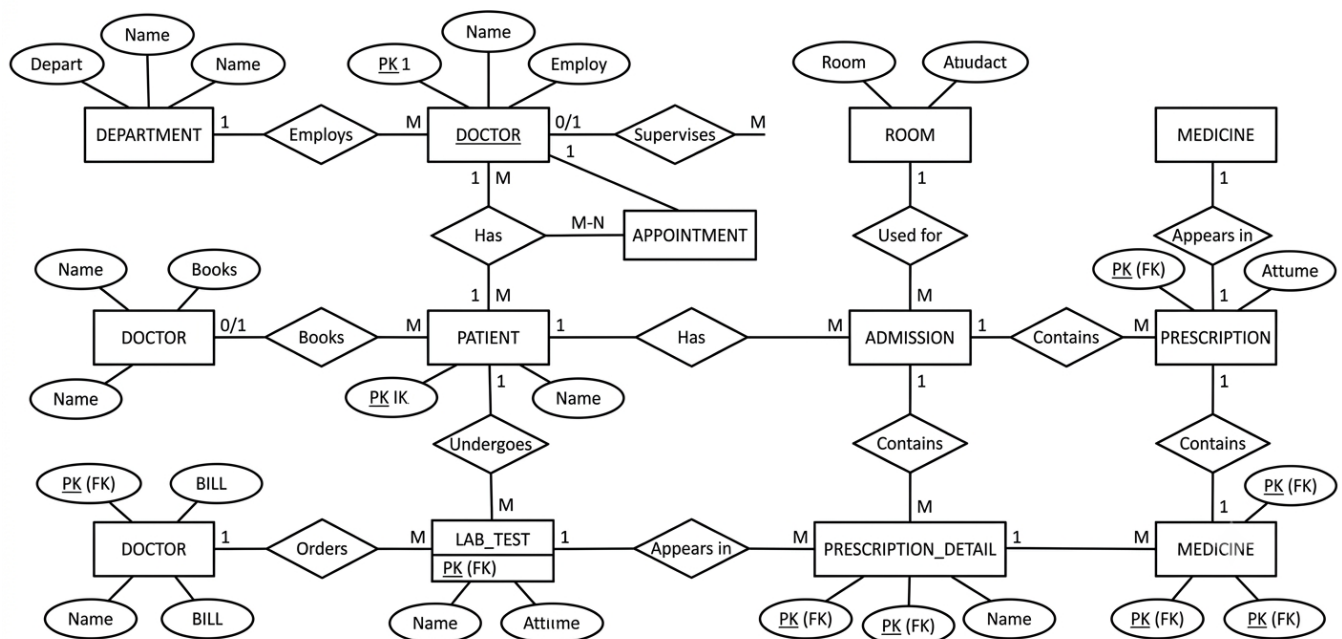
RELATIONSHIPS AND CARDINALITIES

Entity 1	Relationship	Entity 2	Cardinality
DEPARTMENT	Employs	DOCTOR	1:M
DOCTOR	Has	APPOINTMENT	1:M
PATIENT	Books	APPOINTMENT	1:M
PATIENT	Has	ADMISSION	1:M
DOCTOR	Supervises	ADMISSION	1:M
ROOM	Used for	ADMISSION	1:M
APPOINTMENT	Has	PRESCRIPTION	1:M
PRESCRIPTION	Contains	PRESCRIPTION_DETAIL	1:M
MEDICINE	Appears in	PRESCRIPTION_DETAIL	1:M
PATIENT	Undergoes	LAB_TEST	1:M
DOCTOR	Orders	LAB_TEST	1:M
PATIENT	Has	BILL	1:M
APPOINTMENT	Linked to	BILL	1:M
ADMISSION	Linked to	BILL	1:M

ER DIAGRAM

The ER diagram represents the Hospital Management System using the entities, attributes, primary keys, foreign keys, relationships, and cardinalities identified above.

ER DIAGRAM – HOSPITAL MANAGEMENT SYSTEM



RELATIONAL SCHEMA

TABLE	PRIMARY KEY	FOREIGN KEY
DEPARTMENT	department_id	head_doctor_id
DOCTOR	doctor_id	department_id
PATIENT	patient_id	-
APPOINTMENT	appointment_id	patient_id, doctor_id
ROOM	room_id	-
ADMISSION	admission_id	patient_id, room_id, doctor_id
PRESCRIPTION	prescription_id	appointment_id, patient_id, doctor_id
MEDICINE	medicine_id	-
PRESCRIPTION_DETAIL	prescription_detail_id	prescription_id, medicine_id
LAB_TEST	test_id	patient_id, doctor_id
BILL	bill_id	patient_id, appointment_id, admission_id

RELATIONAL SCHEMA WITH ATTRIBUTES

TABLE	RELATIONAL ATTRIBUTES
DEPARTMENT	department_id, department_name, floor_no, phone_no, head_doctor_id
DOCTOR	doctor_id, name, specialization, department_id, phone_no, email, experience_years, consultation_fee
PATIENT	patient_id, name, dob, gender, phone_no, address, blood_group, emergency_contact
APPOINTMENT	appointment_id, patient_id, doctor_id, appointment_date, appointment_time, status, reason
ROOM	room_id, room_number, room_type, floor_no, daily_charge, status
ADMISSION	admission_id, patient_id, room_id, doctor_id, admission_date, discharge_date, admission_reason, status
PRESCRIPTION	prescription_id, appointment_id, patient_id, doctor_id, date_issued, notes
MEDICINE	medicine_id, medicine_name, manufacturer, unit_price, stock_quantity
PRESCRIPTION_DETAIL	prescription_detail_id, prescription_id, medicine_id, dosage, duration_days, quantity
LAB_TEST	test_id, patient_id, doctor_id, test_name, test_date, result, status
BILL	bill_id, patient_id, appointment_id, admission_id, bill_date, total_amount, payment_status, payment_mode

RELATIONSHIP EXPLANATION

Department – Doctor: One department can have many doctors, while each doctor belongs to one department. Therefore, the relationship is 1:M.

Doctor – Appointment: One doctor can have many appointments, while each appointment is booked with one doctor. Therefore, the relationship is 1:M.

Patient – Appointment: One patient can have many appointments over time, while each appointment belongs to one patient. Therefore, the relationship is 1:M.

Patient – Admission: One patient can have many admission records over time, while each admission belongs to one patient. Therefore, the relationship is 1:M.

Doctor – Admission: One doctor can supervise many admissions, while each admission is supervised by one doctor. Therefore, the relationship is 1:M.

Room – Admission: One room can be used for many admissions over time, while each admission is assigned to one room. Therefore, the relationship is 1:M.

Appointment – Prescription: One appointment can have prescription records, while each prescription is linked to one appointment. Therefore, the relationship is 1:M.

Prescription – Prescription_Detail: One prescription can contain many medicine line-items, while each prescription detail belongs to one prescription. Therefore, the relationship is 1:M.

Medicine – Prescription_Detail: One medicine can appear in many prescription detail records, while each prescription detail refers to one medicine. Therefore, the relationship is 1:M.

Patient – Lab_Test: One patient can undergo many lab tests, while each lab test belongs to one patient. Therefore, the relationship is 1:M.

Doctor – Lab_Test: One doctor can order many lab tests, while each lab test is ordered by one doctor. Therefore, the relationship is 1:M.

Patient – Bill: One patient can have many bills over time, while each bill belongs to one patient. Therefore, the relationship is 1:M.

Appointment – Bill: A bill can optionally be linked to an appointment for an outpatient visit. The bill may reference one appointment, while an appointment can be associated with bill records according to the billing design. Therefore, this is represented as 1:M in the relational design.

Admission – Bill: A bill can optionally be linked to an admission for an inpatient stay. The bill may reference one admission, while an admission can be associated with bill records according to the billing design. Therefore, this is represented as 1:M in the relational design.

CONCLUSION

The Hospital Management System has been represented as an Entity Relationship model and converted into a relational schema. The design identifies the major entities, their attributes, primary keys, foreign keys, relationships, and cardinalities required to represent hospital operations in a structured database.